Document Control Report

Title			
Antibiotic guidelines for Gastroenterological indications in Secondary Care			
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0.2	Mar 11	Draft	Change erythromycin to clarithromycin in Campylobacter gastroenteritis. Change to C difficile vancomycin indication.
0.3	May 2011	Draft	Change to references. Change to first line IV antibiotics for biliary infection
0.4	May 2011	Draft	Change to prophylaxis for PEG. Other minor changes to text.
1.0	Jul 2011	Final	Changes to loperamide use and pancreatitis notes after consultation with Antibiotic Working Group. Interim approval was given by the Lead Clinician for Drug and Therapeutics on 26th July 2011.
1.1	Aug 2011	Revision	Minor amendments by Corporate Governance to document control report and header, hyperlinks to appendices.
1.2	Sep 2011	Revision	Minor amendments by DTG.
1.3	Jan 2013	Revision	Change to guidance on management of appendicitis (appendix G) to incorporate guidance on surgical prophylaxis.
1.4	Mar 2013	Revision	Incorporation of guidance on prophylaxis into Antibiotic Guidelines for Surgical Prophylaxis. Change
1.5	Jun 2013	Revision	Minor amendments by Corporate Governance to document control report.

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1 Introduction

This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines for appropriate microbiological investigation and antimicrobial prescribing in adult patients with infection of gastroenterological focus.

2 Purpose

The following general principles can be applied in order to improve the:

- Management of gastroenterological infection in secondary care.
- Setting a standard of care for the management of common secondary care issues (egg. surgical prophylaxis).
- Setting a standard of care for less common diseases to facilitate a consistent approach between gastroenterology, microbiology and pharmacy in terms of patient management, specimen processing and drug availability.

This guideline applies to all adults and must be adhered to. Non compliance with this guideline may be for valid clinical reasons only. The reason for non-compliance must be documented clearly in the patient's notes. This guideline is primarily aimed at all prescribing teams but other staff (e.g. nursing staff, pharmacists) may need to familiarise themselves with some aspects of the guideline.

See separate guidelines for management of gastroenterological conditions in paediatrics.

3 Definitions/ Abbreviations

SBP Spontaneous Bacterial Peritonitis

ERCP Endoscopic Retrograge Cholangio-Pancreatography

PEG Percutaneous Endoscopic Gastrostomy

4 Contact numbers

- Microbiologist Bleep 193. Via switchboard out of hours.
- Antibiotic Pharmacist Bleep 029

5 Investigation, Treatment and Prophylaxis of Gastroenterology infections

See Appendices (hyperlinks below):

Acute gastroenteritis, including traveller's diarrhoea

Acute cholangitis / biliary sepsis

Diverticulitis

Spontaneous bacterial peritonitis

Spontaneous Bacterial Perintonitis - Prophylaxis in variceal bleeding

Clostridium difficile infection

Other intra-abdominal infection (e.g. Appendicitis, perforation)

Pancreatitis

Liver Abscess

Prophylaxis of gastroenterological procedures (including ERCP and PEG insertion)

6 Education and training

Responsibility for education and training lies with the Lead Consultant Microbiologist for Antibiotic Stewardship. It will be provided through formal study days and informal training on the ward.

7 Consultation, Approval, Review and Archiving Processes

The author consulted with all relevant stakeholders. Please refer to the Document Control Report.

Approval was given by the Drug and Therapeutics Committee on 21st March 2013.

The author will be responsible for ensuring the guidelines are reviewed and revisions approved by the Drug and Therapeutics Group in accordance with the Document Control Report.

All versions of these guidelines will be archived in electronic format by the author within the Pharmacy policy archive.

Any revisions to the final document will be recorded on the Document Control Report.

To obtain a copy of the archived guidelines, contact should be made with the author.

8 Monitoring Compliance and Effectiveness

Monitoring of implementation, effectiveness and compliance with these guidelines will be the responsibility of the Lead Clinician for Antibiotic Stewardship. Where non-compliance is found, the reasons for this must have been documented in the patient's medical notes.

Incidents involving gastroenterological infection should be reported according to the Trust's Incident Reporting Policy. Critical incident reports relating to gastroenterological infection will be collated by the Antibiotic Pharmacist. Results will be reported on an annual basis to the Drug and Therapeutics Group.

Suggested audit criteria :

PEG insertion

- Percentage of patients who have had a PEG insertion who have had a wound swab taken in the 30 days after insertion.

- Percentage of patients who have had a PEG insertion who have had a wound swab taken in the 30 days after insertion with isolation of *Staphylococcus aureus* or a haemolytic streptococcus (A, C or G).
- Percentage of patients who have had a PEG insertion who had an MRSA screen in the 6 weeks prior to the procedure

Use of gentamicin is audited by the antibiotic pharmacist on a prospective basis.

Pancreatitis

- At least two sets of blood cultures should be sent before starting antibiotics
- An abdominal ultrasound should be carried out in all patients with pancreatitis within 24 hours

9 References

Solomkin JS et al. (2009) Diagnosis and Management of Complicated Intra-abdominal Infection in Adults and Children: Guidelines by the Surgical Infection Society and the Infectious Diseases Society of America *Clin Infect Dis* **50**, 133-64

Cholangitis

Wada et al. Diagnostic criteria and severity assessment of acute cholangitis: Tokyo Guidelines J Hepatobiliary Pancreat Surg. 2007 January; 14(1): 52–58.

ERCP prophylaxis

British Society of Gastroenterology (2001) Antibiotic prophylaxis in Gastrointestinal Endoscopy. *British Society of Gastroenterology Guidelines.*

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Pancreatitis

Dellinger EP, et al (2007) Early antibiotic treatment for severe acute necrotizing pancreatitis: a randomized, double-blind, placebo-controlled study. *Ann Surg.* **245**(5):674-83.

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van Santvoort HC,et al (2010) A step-up approach or open necrosectomy for necrotizing pancreatitis. *N Engl J Med.* **362**(16):1491-502.

PEG insertion

Blomberg J et al. (2010) Novel approach to antibiotic prophylaxis in percutaneous endoscopic gastrostomy (PEG): randomised controlled trial. *BMJ* **340***,* 3115-3120

Horiuchi A, Nakayama Y, Kajiyama M, Fujii H, Tanaka N. (2006) Nasopharyngeal decolonization of methicillin-resistant Staphylococcus aureus can reduce PEG peristomal wound infection. *Am J Gastroenterol*. **101**, 274-7.

Spontaneous Bacterial Peritonitis

Chavez-Tapia NC et al. (2009) Antibiotics for spontaneous bacterial peritonitis in cirrhotic patients. *Cochrane Database of Systematic Reviews* 2009, Issue 1.

Cohen MJ et al. (2009). Antibiotic prophylaxis for spontaneous bacterial peritonitis in cirrhotic patients with ascites, without gastro-intestinal bleeding. *Cochrane Database of Systematic Reviews 2009, Issue 2.*

Lontos S, Gow PJ, Vaughan RB, Angus PW. (2007). Norfloxacin and trimethoprim-sulfamethoxazole therapy have similar efficacy in prevention of spontaneous bacterial peritonitis. *J Gastroenterol Hepatol.* **23**, 252-5.

Scottish Intercollegiate Guidelines Network. (2008) Management of acute upper and lower gastrointestinal bleeding.

10 Associated Documents

- Incident Reporting Policy
- Antibiotic guidelines for Management of Severe sepsis and Septic Shock

Appendix A – Acute gastroenteritis, including travellers' diarrhoea

Acute gastroenteritis, including travellers' diarrhoea

Oral

1st line	**Conservative management.**
2nd line	**Clarithromycin** 500mg BD PO Consider in persistent diarrhoea caused by *Campylobacter*, or in immunosuppressed patients with *Camplyobacter* and evidence of severe disease.

Help notes

1. Diarrhoea is usually treated with rehydration. Antibiotics are rarely indicated, and may exacerbate symptoms.
2. Viral causes of diarrhoea are common and highly infectious – see Infection Control Guidance on management.
3. E. coli O157 should be considered, especially in bloody diarrhoea. Renal failure occurs in 5% of patients. Antibiotics should not be given.
4. Consider *C. difficile* infection, especially if recent antibiotic prescriptions – see separate guidance
5. Diarrhoea may be a symptom of systemic infection not related to the gastrointestinal tract. Eg. Sepsis and Toxic shock syndrome. Phone a consultant microbiologist to discuss if necessary.
6. Send a stool sample for culture (looks for E. coli O157, Salmonella, Shigella and Campylobacter) and *C. difficile* toxin detection.
7. Request examination for ova, cysts and parasites and Vibrio spp. if there is a travel history to countries where parasitic diseases are common.
8. Anti-motility agents should generally be avoided in infective diarrhoea. This is an absolute contraindication if the patient has bloody diarrhea or is systemically unwell.
9. Traveller's diarrhoea is generally self-limiting and should not be managed differently to other forms of diarrhoea upon return to the UK.

Appendix B – Infections of the Biliary Tract

Infections of the Biliary Tract

Oral treatment

1st line	**Trimethoprim** 200mg BD PO **plus** **Metronidazole** 400mg TDS PO **Usually given for 7 days**

2nd line	**Co-amoxiclav** 625mg TDS PO **Usually given for 7 days**

Intravenous treatment

1st line	**Piperacillin-tazobactam (Tazocin)** 4.5g TDS IV

2ND line / penicillin allergy	**Gentamicin** IV (as per Trust protocol) **plus** **Metronidazole** 500mg TDS IV Consider adding **amoxicillin 1g QDS IV** for enterococcal cover in patients with more severe disease, especially post-ERCP or if there is a stent *in situ.* If penicillin allergy, then consider **vancomycin** (as per Trust protocol) in this situation. **Oral switch.** Use the oral treatment protocol and culture results to determine an appropriate oral antibiotic regime.

Help notes

Biliary colic – *non-inflammatory gall bladder pain (typically gall bladder contraction)*

- RUQ pain, usually due to a stone (10% are acalculous)
- Antibiotics are not indicated

Cholecystitis - *gall bladder inflammation (eg. Fever or raised WBC).*

- Evidence supports the use of early surgery
- Antibiotics are usually indicated
- Oral treatment may be appropriate in patients who are not systemically unwell

Cholangitis – *infection in the biliary tree often in an obstructed system*

- Acute cholangitis is a clinical diagnosis
  - Fever is present in more than 90%
  - Charcot's triad = fever, RUQ pain and jaundice
  - Raised inflammatory markers are usual
  - An abnormal biliary tract is usually seen on imaging
- Antibiotics are indicated in all cases of cholangitis.

- Most patients admitted to hospital with suspected biliary infection will be started on IV antibiotics
- In cases of diagnostic doubt, it may be preferable to give gentamicin plus metronidazole and review the diagnosis after 24 hours.
- Blood cultures should always be taken before treating biliary infection

Red flags

In patients with septic shock – see separate guidelines on management of severe sepsis / septic shock Click here.

Appendix C - Diverticulitis

Diverticulitis

Oral treatment (see notes*)

1st line

Trimethoprim 200mg BD PO

plus

Metronidazole 400mg TDS PO

Usually given for 7 days

2nd line

Co-amoxiclav 625mg TDS PO

Usually given for 7 days

Intravenous treatment

Gentamicin IV

plus

Metronidazole 500mg TDS IV

Consider adding **amoxicillin 1g QDS IV** for enterococcal cover in patients with more severe disease. If penicillin allergy, then consider **vancomycin** (as per Trust protocol) in this situation.

Oral switch. Use the oral treatment protocol and culture results to determine an appropriate oral antibiotic regime.

*** Antibiotic choices**

There is no evidence to prefer one regimen over another. Co-trimoxazole and co-amoxiclav are probably equally effective and are recommended by some guidelines. Experience with trimethoprim is less, but it has similar activity to co-trimoxazole and has a better side effect profile than co-trimoxazole, co-amoxiclav or ciprofloxacin. In particular, the risk of *C. difficile* is significantly lower with trimethoprim. Previous urine culture results may help inform antibiotic choices.

Help notes

- *Diverticulosis*. (An incidental finding of diverticula, asymptomatic by definition)
- *Diverticular disease.* (Intermittent lower abdo pain caused by diverticula)
  - Management involves diet, paracetamol, consider trial of Rx for IBS. Antibiotics are not indicated.
- *Diverticulitis*. (Inflammation and infection of diverticula causing marked abdo pain, often with fever.)

Diagnosis of diverticulitis is based on signs and symptoms

- Constant abdo pain, often severe with fever and altered bowel habit
- Raised inflammatory markers
- Confirmed presence of diverticula
- Excluding differentials (eg. IBS, appendicitis, colitis, malignancy, UTI, PID, ectopic pregnancy)

Management of diverticulitis in the community

- Mild diverticulitis can be managed at home with clear fluid, analgesia and oral antibiotics
- Gradually reintroduce solid food as symptoms improve.
- Should improve within 48h

Management of diverticulitis in hospital

- Usually intravenous antibiotics, with clear fluid and analgesia.
- Diagnosis is usually confirmed with CT, which also looks for complications.
- Further investigations (including imaging) must be carried out in patients not improving after 48h.

- *Complicated diverticulitis.* (Diverticulitis with formation of abscesses, perforation or fistulae)
  - Manage as for diverticulitis for smaller abscesses (less than 4cm)
  - For larger abscess consider percutaneous drainage or surgical management
  - If patient is not improving then phone Microbiology for advice.

Red flags

- Admit if :
  - Frail
  - Dehydrated / poor fluid intake
  - Significant bleeding
  - Suspected complication
  - Failure to control symptoms after 48h
- Consider perforation in patients who show signs of severe sepsis.
  - See severe sepsis protocol

Appendix D – Spontaneous Bacterial Peritonitis

Spontaneous Bacterial Peritonitis

Treatment

1st line	**Piperacillin-tazobactam (Tazocin)** 4.5g TDS IV
Penicillin allergy	**Ciprofloxacin** 500mg BD PO or 400mg BD IV Only IV if unable to take oral medication. **plus** **Vancomycin** IV dose according to Trust protocol Covers enterococci. Do not give further doses unless supported by positive cultures.

Organisms and sensitivity data

		Trim	Aug	Cip	Gent	Cef
Common	*E. coli (80%)*	87% (2006-8)	62%	88%	100%	80%
Less common	Faecal streptococci (10%)					
	Polymicrobial infection (10%)					
Rare	*Pseudomonas* spp,					
	Anaerobes					
Very rare	Mycobacteria					
	Fungi					

Help notes

BEFORE STARTING ANTIBIOTICS :

- Ascitic fluid (10ml) should be inoculated into aerobic and anaerobic blood culture bottles
- Ascitic fluid should be sent in an EDTA blood tube for a cell count (haematology)
- Take blood and urine cultures.

1. An ascitic neutrophil count of more than 250 cells / ml has more than 90% sensitivity and specificity for diagnosis of SBP.
2. Patients with raised ascitic neutrophils, but negative culture, should be treated as having SBP.
3. Patients with positive culture but without a rise in ascitic neutrophils should be kept under close observation, and a repeat ascitic tap considered.
4. Malignancy, tuberculosis and fungal infections should be considered in patients with raised ascitic lymphocytes.

Duration and oral switch

- Generally, SBP is treated for a minimum of 5 days, but antibiotics should be stopped when there is clinical resolution of symptoms and a repeat ascitic tap has shown a fall in the ascitic neutrophil count to below 250 neutrophils/ml.
- Oral switch to co-amoxiclav (625mg TDS) or trimethoprim (200mg BD), according to culture results

Red flags

SBP should always be suspected in patients with possible infection and ascites. The following symptoms and signs may be present

- Fever and / or rigors in most patients
- Abdominal pain and discomfort in most patients
- Worsening encephalopathy, diarrhoea, diuretic resistant ascites, worsening renal function, ileus
- Many infections may be poorly localized.

In severe sepsis (ie. Evidence of end organ dysfunction) consider using the severe sepsis algorithm.

Appendix E – Prophylaxis for Spontaneous Bacterial Peritonitis

Prophylaxis for Spontaneous Bacterial Peritonitis

Including in variceal bleeding

Prophylaxis in variceal bleeding

1st line	**Piperacillin-tazobactam (Tazocin)** 4.5g TDS IV

Penicillin allergy	**Trimethoprim** 200mg BD PO In low risk patients (eg Minimal ascites, normal serum bilirubin) **or** **Ciprofloxacin** 500mg BD PO or 400mg BD IV In higher risk patients, but balance against increased risk of *C. difficile* infection. Use intravenous route only if oral therapy not possible.

Organisms and sensitivity data (2008-10)

		Trim	Aug	Cip	Gent	Cef
Common	*E. coli (80%)*	87% (2006-8)	62%	88%	100%	80%
Less common	Faecal streptococci (10%)					
	Polymicrobial infection (10%)					
Rare	*Pseudomonas* spp,					
	Anaerobes					
Very rare	Mycobacteria					
	Fungi					

Help notes

BEFORE STARTING ANTIBIOTICS :

- Ascitic fluid (10ml) should be inoculated into aerobic and anaerobic blood culture bottles
- Ascitic fluid should be sent in an EDTA blood tube for a cell count (haematology)
  See SBP protocol for advice on interpretation of cell counts.

Prophylaxis in variceal bleeding

1. There is good evidence that patients with variceal bleeding should receive antibiotic prophylaxis.
2. An attempt should be made to obtain a specimen of ascitic fluid for culture and cell count before starting antibiotics.
3. The optimum duration of prophylactic antibiotics after a variceal bleed is poorly defined.
4. Patients should be reviewed after a maximum of 48 hours, with the results of the ascitic culture and cell count (if available) and according to clinical progress, with a view to stopping antibiotics in the absence of an ongoing indication.
5. If antibiotics are to be continued, consider switching to trimethoprim (if a trimethoprim sensitive isolate has been recovered, or if cultures are negative).

Long term prophylaxis

1. Patients with ascites are at risk of developing SBP, regardless of aetiology. The risk is highest in those patients with previous SBP, low ascitic albumin and high serum bilirubin. The benefits of antibiotic prophylaxis in these patients remain unclear, and any potential for reduced hospitalisation must be offset against the increased risk of *C. difficile* and development of antibiotic resistance.

2. Long-term antibiotic prophylaxis should be tailored to the resistance profile of previous isolates. Suitable first line agents may include trimethoprim (200mg BD) or ciprofloxacin (500mg BD).

Appendix F – Clostridium difficile infection

Clostridium difficile infection

Oral treatment

1st line	**Metronidazole** 500mg TDS IV **or** **Metronidazole** 400mg TDS PO In confirmed disease, always treat for a minimum of 10 days, regardless of symptom resolution.
2nd line – **or 1st line in severe disease**	**Vancomycin** 125mg QDS PO In confirmed disease, always treat for a minimum of 10 days, regardless of symptom resolution.

Help notes

See *C. difficile* policy for full details on management of infection.

Assess disease severity :

- WBC more than15
- Acute rise in creatinine (more than150% above baseline)
- Temp more than 38°C
- Evidence of toxic megacolon / pseudomebranous colitis

Stop other antibiotics if possible. If considered necessary then discuss with a consultant microbiologist.

Stop proton pump inhibitors and laxatives if possible.

Review to stop codeine and loperamide, especially if severe disease.

Consider empirical treatment while toxin results are pending if high index of suspicion (eg. Severe diarrhoea with recent use of high risk antibiotics (quinolones, cephalosporins, co-amoxiclav, clindamycin) and patient has markers of severe infection.)

If antibiotics are started empirically, then continued use must be reviewed with the toxin result (tested 7 days per week)

All patients with suspected *C. difficile* infection must be isolated.

Appendix G – Appendicitis and Other intra-abdominal infection

Appendicitis
and other intra-abdominal infection eg. perforation

Help notes

Intravenous treatment

Gentamicin IV as per protocol (5mg/kg)

plus

Metronidazole 500mg TDS IV

In hospital acquired infection (eg. Post –operative infection) and more severe disease then consider adding **amoxicillin** 1g QDS IV. Phone to discuss if penicillin allergy.

Oral switch. Use the oral treatment protocol for diverticulitis and culture results to determine an appropriate oral antibiotic regime.

Specific guidance on management of Appendicitis

- **Give antibiotics (as above) on diagnosis** of appendicitis
- Give standard pre-operative antibiotics before surgery, unless less than 1 hour since administration of treatment dose antibiotics on ward.

- In patients undergoing an exploratory procedure when diagnosis uncertain
  - do not give antibiotics pre-operatively
  - If appendicitis is diagnosed during surgery and prophylactic antibiotics have not been given, then **give treatment dose antibiotics in theatre** (as above)

24 hours of antibiotics (ie. A single treatment dose of gentamicin and 3 doses of metronidazole) are sufficient if no evidence of gangrene, perforation or peritonitis.

Duration of antibiotics less than 24 hours if :

- Bowel injury due to penetrating, blunt or iatrogenic trauma that are repaired within 12 hours with no
- Perforation of stomach, duodenum, and proximal jejunum
- Acute appenditicits without evidence of gangrene, perforation, or peritonitis

Duration in other situations should be determined by clinical response. Antibiotics can be stopped when :

1. WBC less than 75% of peak
2. Afebrile for 24 hours
3. Normal bowel function
4. No inotropic support

If antibiotics are considered necessary for more than 5 days, then there must be consideration of an alternative diagnosis or intraabdominal collection.

Patients with intra-abdominal infection often have non-specific chest signs with opacity at the lung bases on CXR.

- Be wary of a diagnosis of pneumonia in these patients.

Anti-fungal therapy is generally not indicated unless yeasts have been isolated and the patient is not responding to empirical antibiotic therapy. Use **fluconazole 400mg OD PO.**

Red flags

- Consider perforation in patients who show signs of severe sepsis
- Consider septic shock if not responding to aggressive resuscitation.
  - See severe sepsis and septic shock protocol

Appendix H - Pancreatitis

Pancreatitis

Antibiotic prophylaxis is discouraged

Intravenous treatment of secondary infection

(preferably after percutaneous drainage)

Meropenem 500mg QDS IV

Consider addition of antifungal agents

(eg. **Fluconazole** 400mg OD

in patients not responding to first line treatment or if multiple isolates of yeasts from clinical specimens.

Help notes

Acute pancreatitis is a non-infective condition caused by derangement of pancreatic enzyme regulation.

- Pyrexia is common in early pancreatitis and does not indicate infection.
- Secondary infection is uncommon in the first week after onset.
- Prophylactic antibiotics do not reduce the incidence of infection, and infecting organisms tend to be more resistant after use of prophylaxis.
- CRP is often raised as a consequence of inflammation, although a secondary rise after a few days may be a marker of infection.
- Infection is rare if necrosis does not exceed 30%.
- The gold standard for diagnosis of infection remains culture of organisms from infected sites, taken either by radiological intervention or at surgery.
- An abdominal ultrasound should be carried out in all patients with pancreatitis to look for biliary pathology.
- It is unlikely that secondary infection will resolve with antibiotics alone.
    - Radiological drainage or open necrosectomy must be considered

1. All patients who are pyrexial should have two sets of blood cultures sent.
2. If antibiotics are started early, then serious consideration should be given to stopping if necrosis is less than 30% on contrast-enhanced CT (if performed more than 48 hours after onset of symptoms).
3. In patients for whom infection is considered likely a radiologically guided aspiration of tissue should be attempted prior to starting antibiotic therapy.

Red flags

- Consider septic shock if not responding to aggressive resuscitation.

See severe sepsis and septic shock protocol

Appendix I – Liver abscess

Liver abscess

Initial treatment depends on likely focus and previous antibiotics.

Strongly consider withholding antibiotics pending aspiration.

Discuss with microbiology if empirical antibiotics required.

Help notes

Pyogenic liver abscesses commonly arise from

- Infections of the biliary tract
- Via the portal vein in appendicitis / diverticulitis / colonic malignancy
- Via the hepatic vein in endocarditis

Candida liver abscesses may occur in severely immunocompromised patients.

Consider amoebic liver abscess, especially if travel history to endemic areas.

- Abscess is usually solitary and affects the right lobe.
- Serology is sensitive.
- Treatment usually conservative with metronidazole and a luminal amoebicide.

Consider hydatid disease (caused by Echinococcus).

- USS / CT scan is diagnostic and serology is confirmatory. Phone for advice.
- Avoid needle aspiration

Blood cultures should be sent in all patients with suspected liver abscess

Strongly consider withholding antibiotics before aspiration of an abscess as culture results will be used to guide long-term management.

In abscesses less than 3cm, antibiotic management alone (minimum 6 weeks) may be appropriate.

In larger abscesses then drainage must be attempted unless there are clear contraindications.

Oral antibiotics may be appropriate when the patient has stabilized, although prolonged IV treatment may be required in large collections that cannot be drained. Discuss with microbiology.

Red flags

- Consider septic shock if not responding to aggressive resuscitation.
  - See severe sepsis and septic shock protocol

Appendix J – Prophylaxis of gastroenterological procedures

Prophylaxis of gastroenterological procedures (including ERCP and PEG insertion)

Guidance can be found in the document "Antibiotic Guidelines for Surgical Prophylaxis."